

AMITABH KUMAR

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EDUCATION

University of New South Wales (UNSW)

Bachelor of Software Engineering (Honours)

Kensington, NSW

Feb 2023 - Dec 2026

[Letter of Recommendation from Prof. Fethi Rabhi, Director of Software Engineering, UNSW](#)

WORK EXPERIENCE

Bloomberg L.P. x FAIC

AI Research Intern

Jan 2026 - Present

- First authored paper - “*VOTR: Vector Orchestrated Tool Retrieval for Scalable Multi-Agent Systems*”
- Designed and implemented a vector-based tool retrieval system that replaces rule-based and sequential tool routing, attaining Top 1 retrieval of 96.8% on a dataset of 309 MCPs and 2806 tools with sub 350ms latency paired with a LangChain agent to orchestrate a production ready interface.
- Built and evaluated end-to-end multi-agent architectures integrating LLMs, vector search and tool execution, benchmarking performance across latency, accuracy and system throughput.
- Conducted empirical experiments and ablation studies to analyse trade-offs between retrieval strategies, contributing to system design decisions and research findings.

University of New South Wales

Academic Tutor

Jan 2026 - Present

- Delivering weekly workshops for an industry partnered course (AWS) focused on fullstack system design, cloud native architecture and scalable software development.
- Guiding students through practical use of cloud services for authentication, databases, serverless compute, VPC, containerization and CI/CD as part of a trimester long fullstack project.

Fintech AI Innovation Consortium (FAIC)

Software Engineering Intern

April 2025 – Dec 2025

- Collaborating with PhD researchers to lead database refactoring, AI/NLP experimentation and fullstack development for automated business process systems.
- Designed and implemented a secure, scalable, PEPPOL-compliant e-invoicing API using AWS Lambda with schema validation aligned to regulatory standards.
- Contributing to development of an automation system using AWS RDS, API Gateway and Lambda for APIs, Elastic Beanstalk for frontend hosting and Step Functions to orchestrate AI agentic workflows.

InspireEd (Acquired Family Business)

Product/Operations

2023 – July 2025

- Led product and operations for a training and consulting firm, owning GTM strategy, internal tooling and data-driven decision making across B2B and B2C offerings.
- Designed and implemented internal systems for client onboarding and automated notifications, improving operational efficiency and scalability.

RESEARCH & PUBLICATIONS

- (First author) VOTR: Vector Orchestrated Tool Retrieval for Scalable Multi-Agent Systems [Authors: A.Kumar, F.Rabhi, E.Wang]
- (ICEIS 2026) Enterprise Information Exchange via IE-BSE Framework. [Authors: M.R.Raza, F.Rabhi, G.Joukadar, A.Kumar]
- Member of Fintech AI Innovation Consortium ([FAIC](#)) participating in research, development and engagements with industry partners.
- Member of Algorithmic Decision Theory ([ADT](#)) groups at UNSW participating in weekly readings.
- Member of Agentic AI Foundation ([AAIF](#)) participating in online discussions with engineers from Anthropic, HuggingFace, Microsoft and more to help shape the future of the MCP framework.

PROJECTS

VOTR: Vector Orchestrated Tool Retrieval

2026

Python, TypeScript, LLMs, LangChain, Vector Databases, Multi-Agent Systems

- Created a routing vector-based retrieval paradigm for scalable tool selection in multi-agent systems.
- Built [VOTR Orchestrator](#), a LangChain agent enabling dynamic routing of agent queries to tools and APIs via semantic similarity instead of rule-based pipelines utilising the VOTR routing system.
- Designed and evaluated multi-agent workflows, improving latency and scalability of tool execution through embedding-driven orchestration for up to 309 MCPs + 2807 tools.

Full-Stack UBL Invoice Automation Platform

Jan - April 2025

TypeScript, Node.js, Express, AWS (RDS, S3, Elastic Beanstalk), Gemini AI, Vercel

- Built a full-stack platform for UBL invoice generation, validation, translation and PDF delivery using REST APIs and AI-assisted workflows.
- Implemented schema validation, compliance checks and PostgreSQL persistence on AWS RDS with CI/CD pipelines via GitHub Actions.

Open-Source NPM Package

March 2024

TypeScript, JavaScript, NPM

- Published a package called `any-call-system-web` on NPMjs, a flexible call routing and trigger library used by multiple projects, achieving 100+ downloads.

EXTRACURRICULARS & AWARDS

- [Recipient of a Letter of Recommendation from Prof. Fethi Rabhi](#) recognising technical and academic distinction.
- [Selected for STaR program from Celonis](#) in support of my final year thesis at UNSW in data, AI and automation discovery.
- Top 10% Achiever in the Australian Intermediate Mathematics Olympiad (AIMO).
- [Certificate of Achievement in Financial Literacy with Models](#) (DKM & ClayMatter).
- Certificates from Bloomberg for [Bloomberg Market Concepts](#), [Bloomberg Finance Fundamentals](#), [Bloomberg Spreadsheet Analysis](#), and [Environmental Social Governance](#).
- [First Place — Most Aesthetic Design Hackathon](#); engineered a Mars Rover with bidirectional movement, remote-controlled scoop and real-time telemetry.
- [Volunteer Lead Tutor at Code Club Australia](#); mentored high school students in foundational programming.
- Volunteer Training Lead at UNSW DevSoc; mentored first year computer science and engineering students on software fundamentals.

REFERENCES

Available upon request